symembed: graded packing of simple cases, written out in full

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Each section shows the graded embedding (`embedder="bursar_graded"`, notes Section 3.4) of one case: the chosen h (order r) and s, the invariants output directly, the packed coordinates z_j (exact complex linear forms in the d's and e's, at power x^j, with their h-weights), and every output expanded in the d's and e's with exact coefficients. Expansions with more than 100 terms show only their term count; all appear in full in `simple_cases_graded_full.txt`. Generated by `symembed/explore/simple_cases_graded.py`; do not edit by hand.

1 $S(2)$ acting on (a, b), $k = 2$

$|G| = 2$, $m = 4$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 4 (plain Bursarial: 11; quotient dimension 3).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{ab}$

j	weight	real part of z_j	imaginary part of z_j
0	1	$d_{aa} - d_{bb}$	$2e_{ab}$

output	expanded
$inv[0]$	$d_{aa} + d_{bb}$
$inv[1]$	$2d_{ab}$
$ReE_{2,0}$	$-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + 4e_{ab}^2$
$ImE_{2,0}$	$-4d_{aa}e_{ab} + 4d_{bb}e_{ab}$

2 $S(3)$ acting on (a, b, c), $k = 2$

$|G| = 6$, $m = 9$ slots; $h = (abc)$ (order $r = 3$), $s = (bc)$, mode x; real coefficients. Outputs: 32 (plain Bursarial: 174; quotient dimension 5).

j	weight	real part of z_j	imaginary part of z_j
0	0	$2d_{aa} + 2d_{bb} + 2d_{cc}$	$2e_{ab} - 2e_{ac} + 2e_{bc}$
1	1	$2d_{aa} - d_{bb} - d_{cc}$	$\sqrt{3}\,(-d_{bb} + d_{cc})$
2	2	$\frac{d_{ab}}{2} + \frac{d_{ac}}{2} - d_{bc}$	$\frac{\sqrt{3}(d_{ab} - d_{ac})}{2}$
3	0	$2d_{ab} + 2d_{ac} + 2d_{bc}$	0
4	1	$\frac{3e_{ab}}{2} + \frac{3e_{ac}}{2}$	$\frac{\sqrt{3}(e_{ab} - e_{ac} - 2e_{bc})}{2}$

output	expanded
$ReE_{1,0}$	$12d_{aa} + 12d_{bb} + 12d_{cc}$
$ReE_{1,3}$	$12d_{ab} + 12d_{ac} + 12d_{bc}$
$ReE_{2,0}$	$60d_{aa}^2 + 120d_{aa}d_{bb} + 120d_{aa}d_{cc} + 60d_{bb}^2 + 120d_{bb}d_{cc} + 60d_{cc}^2 + 12e_{ab}^2 - 24e_{ab}e_{ac} + 24e_{ab}e_{bc} + 12e_{ac}^2 - 24e_{ac}e_{bc} + 12e_{bc}^2$
$ReE_{2,3}$	$114d_{aa}d_{ab} + 114d_{aa}d_{ac} + 132d_{aa}d_{bc} + 114d_{ab}d_{bb} + 132d_{ab}d_{cc} + 132d_{ac}d_{bb} + 114d_{ac}d_{cc} + 114d_{bb}d_{bc} + 114d_{bc}d_{cc}$
$ReE_{2,6}$	$60d_{ab}^2 + 120d_{ab}d_{ac} + 120d_{ab}d_{bc} - 9d_{ab}e_{ac} - 9d_{ab}e_{bc} + 60d_{ac}^2 + 120d_{ac}d_{bc} - 9d_{ac}e_{ab} + 9d_{ac}e_{bc} + 60d_{bc}^2 + 9d_{bc}e_{ab} + 9d_{bc}e_{ac}$
$ReE_{3,0}$	$160d_{aa}^3 + 480d_{aa}^2d_{bb} + 480d_{aa}^2d_{cc} + 480d_{aa}d_{bb}^2 + 960d_{aa}d_{bb}d_{cc} + 480d_{aa}d_{cc}^2 + 96d_{aa}e_{ab}^2 - 192d_{aa}e_{ab}e_{ac} + 192d_{aa}e_{ab}e_{bc} + 96d_{aa}e_{ac}^2 - 192d_{aa}e_{ac}e_{bc} + 96d_{aa}e_{bc}^2 + 160d_{bb}^3 + 480d_{bb}^2d_{cc} + 480d_{bb}d_{cc}^2 + 96d_{bb}e_{ab}^2 - 192d_{bb}e_{ab}e_{ac} + 192d_{bb}e_{ab}e_{bc} + 96d_{bb}e_{ac}^2 - 192d_{bb}e_{ac}e_{bc} + 96d_{bb}e_{bc}^2 + 160d_{cc}^3 + 96d_{cc}e_{ab}^2 - 192d_{cc}e_{ab}e_{ac} + 192d_{cc}e_{ab}e_{bc} + 96d_{cc}e_{ac}^2 - 192d_{cc}e_{ac}e_{bc} + 96d_{cc}e_{bc}^2$
$ReE_{3,3}$	$16d_{aa}^3 + 432d_{aa}^2d_{ab} + 432d_{aa}^2d_{ac} - 24d_{aa}^2d_{bb} + 576d_{aa}^2d_{bc} - 24d_{aa}^2d_{cc} + 864d_{aa}d_{ab}d_{bb} + 1008d_{aa}d_{ab}d_{cc} - 24\sqrt{3}d_{aa}d_{ab}e_{ab} + 24\sqrt{3}d_{aa}d_{ab}e_{ac} - 24\sqrt{3}d_{aa}d_{ab}e_{bc} + 1008d_{aa}d_{ac}d_{bb} + 864d_{aa}d_{ac}d_{cc} + 24\sqrt{3}d_{aa}d_{ac}e_{ab} - 24\sqrt{3}d_{aa}d_{ac}e_{ac} + 24\sqrt{3}d_{aa}d_{ac}e_{bc} - 24d_{aa}d_{bb}^2 + 1008d_{aa}d_{bb}d_{bc} + 96d_{aa}d_{bb}d_{cc} + 1008d_{aa}d_{bc}d_{cc} - 24d_{aa}d_{cc}^2 + 432d_{ab}d_{bb}^2 + 1008d_{ab}d_{bb}d_{cc} + 24\sqrt{3}d_{ab}d_{bb}e_{ab} - 24\sqrt{3}d_{ab}d_{bb}e_{ac} + 24\sqrt{3}d_{ab}d_{bb}e_{bc} + 576d_{ab}d_{cc}^2 + 96d_{ab}e_{ab}^2 - 192d_{ab}e_{ab}e_{ac} + 192d_{ab}e_{ab}e_{bc} + 96d_{ab}e_{ac}^2 - 192d_{ab}e_{ac}e_{bc} + 96d_{ab}e_{bc}^2 + 576d_{ac}d_{bb}^2 + 1008d_{ac}d_{bb}d_{cc} + 432d_{ac}d_{cc}^2 - 24\sqrt{3}d_{ac}d_{cc}e_{ab} + 24\sqrt{3}d_{ac}d_{cc}e_{ac} - 24\sqrt{3}d_{ac}d_{cc}e_{bc} + 96d_{ac}e_{ab}^2 - 192d_{ac}e_{ab}e_{ac} + 192d_{ac}e_{ab}e_{bc} + 96d_{ac}e_{ac}^2 - 192d_{ac}e_{ac}e_{bc} + 96d_{ac}e_{bc}^2 + 16d_{bb}^3 + 432d_{bb}^2d_{bc} - 24d_{bb}^2d_{cc} + 864d_{bb}d_{bc}d_{cc} - 24\sqrt{3}d_{bb}d_{bc}e_{ab} + 24\sqrt{3}d_{bb}d_{bc}e_{ac} - 24\sqrt{3}d_{bb}d_{bc}e_{bc} - 24d_{bb}d_{cc}^2 + 432d_{bc}d_{cc}^2 + 24\sqrt{3}d_{bc}d_{cc}e_{ab} - 24\sqrt{3}d_{bc}d_{cc}e_{ac} + 24\sqrt{3}d_{bc}d_{cc}e_{bc} + 96d_{bc}e_{ab}^2 - 192d_{bc}e_{ab}e_{ac} + 192d_{bc}e_{ab}e_{bc} + 96d_{bc}e_{ac}^2 - 192d_{bc}e_{ac}e_{bc} + 96d_{bc}e_{bc}^2 + 16d_{cc}^3$
$ReE_{3,6}$	$36d_{aa}^2e_{ab} + 36d_{aa}^2e_{ac} + 432d_{aa}d_{ab}^2 + 864d_{aa}d_{ab}d_{ac} + 1008d_{aa}d_{ab}d_{bc} - 72d_{aa}d_{ab}e_{ac} - 72d_{aa}d_{ab}e_{bc} + 432d_{aa}d_{ac}^2 + 1008d_{aa}d_{ac}d_{bc} - 72d_{aa}d_{ac}e_{ab} + 72d_{aa}d_{ac}e_{bc} - 72d_{aa}d_{bb}e_{ac} - 72d_{aa}d_{bb}e_{bc} + 576d_{aa}d_{bc}^2 + 72d_{aa}d_{bc}e_{ab} + 72d_{aa}d_{bc}e_{ac} - 72d_{aa}d_{cc}e_{ab} + 72d_{aa}d_{cc}e_{bc} - 2d_{ab}^3 + 3d_{ab}^2d_{ac} + 432d_{ab}^2d_{bb} + 3d_{ab}^2d_{bc} + 576d_{ab}^2d_{cc} + 3d_{ab}d_{ac}^2 + 1008d_{ab}d_{ac}d_{bb} - 12d_{ab}d_{ac}d_{bc} + 1008d_{ab}d_{ac}d_{cc} + 864d_{ab}d_{bb}d_{bc} - 72d_{ab}d_{bb}e_{ac} - 72d_{ab}d_{bb}e_{bc} + 3d_{ab}d_{bc}^2 + 1008d_{ab}d_{bc}d_{cc} - 72d_{ab}d_{cc}e_{ac} - 72d_{ab}d_{cc}e_{bc} - 24\sqrt{3}d_{ab}e_{ab}^2 + 12\sqrt{3}d_{ab}e_{ab}e_{ac} - 12\sqrt{3}d_{ab}e_{ab}e_{bc} + 12\sqrt{3}d_{ab}e_{ac}^2 - 24\sqrt{3}d_{ab}e_{ac}e_{bc} + 12\sqrt{3}d_{ab}e_{bc}^2 - 2d_{ac}^3 + 576d_{ac}^2d_{bb} + 3d_{ac}^2d_{bc} + 432d_{ac}^2d_{cc} + 1008d_{ac}d_{bb}d_{bc} - 72d_{ac}d_{bb}e_{ab} + 72d_{ac}d_{bb}e_{bc} + 3d_{ac}d_{bc}^2 + 864d_{ac}d_{bc}d_{cc} - 72d_{ac}d_{cc}e_{ab} + 72d_{ac}d_{cc}e_{bc} + 12\sqrt{3}d_{ac}e_{ab}^2 + 12\sqrt{3}d_{ac}e_{ab}e_{ac} + 24\sqrt{3}d_{ac}e_{ab}e_{bc} - 24\sqrt{3}d_{ac}e_{ac}^2 + 12\sqrt{3}d_{ac}e_{ac}e_{bc} + 12\sqrt{3}d_{ac}e_{bc}^2 - 36d_{bb}^2e_{ab} + 36d_{bb}^2e_{bc} + 432d_{bb}d_{bc}^2 + 72d_{bb}d_{bc}e_{ab} + 72d_{bb}d_{bc}e_{ac} + 72d_{bb}d_{cc}e_{ab} + 72d_{bb}d_{cc}e_{ac} - 2d_{bc}^3 + 432d_{bc}^2d_{cc} + 72d_{bc}d_{cc}e_{ab} + 72d_{bc}d_{cc}e_{ac} + 12\sqrt{3}d_{bc}e_{ab}^2 - 24\sqrt{3}d_{bc}e_{ab}e_{ac} - 12\sqrt{3}d_{bc}e_{ab}e_{bc} + 12\sqrt{3}d_{bc}e_{ac}^2 + 12\sqrt{3}d_{bc}e_{ac}e_{bc} - 24\sqrt{3}d_{bc}e_{bc}^2 - 36d_{cc}^2e_{ac} - 36d_{cc}^2e_{bc}$
$ReE_{3,9}$	$18d_{aa}e_{ab}^2 + 72d_{aa}e_{ab}e_{ac} + 36d_{aa}e_{ab}e_{bc} + 18d_{aa}e_{ac}^2 - 36d_{aa}e_{ac}e_{bc} - 36d_{aa}e_{bc}^2 + 160d_{ab}^3 + 480d_{ab}^2d_{ac} + 480d_{ab}^2d_{bc} - 72d_{ab}^2e_{ac} - 72d_{ab}^2e_{bc} + 480d_{ab}d_{ac}^2 + 960d_{ab}d_{ac}d_{bc} - 72d_{ab}d_{ac}e_{ab} - 72d_{ab}d_{ac}e_{ac} + 480d_{ab}d_{bc}^2 + 72d_{ab}d_{bc}e_{ab} - 72d_{ab}d_{bc}e_{bc} + 160d_{ac}^3 + 480d_{ac}^2d_{bc} - 72d_{ac}^2e_{ab} + 72d_{ac}^2e_{bc} + 480d_{ac}d_{bc}^2 + 72d_{ac}d_{bc}e_{ac} + 72d_{ac}d_{bc}e_{bc} + 18d_{bb}e_{ab}^2 - 36d_{bb}e_{ab}e_{ac} - 72d_{bb}e_{ab}e_{bc} - 36d_{bb}e_{ac}^2 - 36d_{bb}e_{ac}e_{bc} + 18d_{bb}e_{bc}^2 + 160d_{bc}^3 + 72d_{bc}^2e_{ab} + 72d_{bc}^2e_{ac} - 36d_{cc}e_{ab}^2 - 36d_{cc}e_{ab}e_{ac} + 36d_{cc}e_{ab}e_{bc} + 18d_{cc}e_{ac}^2 + 72d_{cc}e_{ac}e_{bc} + 18d_{cc}e_{bc}^2$
$ReE_{3,12}$	$27e_{ab}^2e_{ac} + 27e_{ab}^2e_{bc} + 27e_{ab}e_{ac}^2 - 27e_{ab}e_{bc}^2 - 27e_{ac}^2e_{bc} - 27e_{ac}e_{bc}^2$

$ReE_{4,0}$ $240d_{aa}^4 + 960d_{aa}^3d_{bb} + 960d_{aa}^3d_{cc} + 1440d_{aa}^2d_{bb}^2 + 2880d_{aa}^2d_{bb}d_{cc} + 1440d_{aa}^2d_{cc}^2 + 288d_{aa}^2e_{ab}^2 - 576d_{aa}^2e_{ab}e_{ac} + 576d_{aa}^2e_{ab}e_{bc} + 288d_{aa}^2e_{ac}^2 - 576d_{aa}^2e_{ac}e_{bc} + 288d_{aa}^2e_{bc}^2 + 960d_{aa}d_{bb}^3 + 2880d_{aa}d_{bb}^2d_{cc} + 2880d_{aa}d_{bb}d_{cc}^2 + 576d_{aa}d_{bb}e_{ab}^2 - 1152d_{aa}d_{bb}e_{ab}e_{ac} + 1152d_{aa}d_{bb}e_{ab}e_{bc} + 576d_{aa}d_{bb}e_{ac}^2 - 1152d_{aa}d_{bb}e_{ac}e_{bc} + 576d_{aa}d_{bb}e_{bc}^2 + 960d_{aa}d_{cc}^3 + 576d_{aa}d_{cc}e_{ab}^2 - 1152d_{aa}d_{cc}e_{ab}e_{ac} + 1152d_{aa}d_{cc}e_{ab}e_{bc} + 576d_{aa}d_{cc}e_{ac}^2 - 1152d_{aa}d_{cc}e_{ac}e_{bc} + 576d_{aa}d_{cc}e_{bc}^2 + 240d_{bb}^4 + 960d_{bb}^3d_{cc} + 1440d_{bb}^2d_{cc}^2 + 288d_{bb}^2e_{ab}^2 - 576d_{bb}^2e_{ab}e_{ac} + 576d_{bb}^2e_{ab}e_{bc} + 288d_{bb}^2e_{ac}^2 - 576d_{bb}^2e_{ac}e_{bc} + 288d_{bb}^2e_{bc}^2 + 960d_{bb}d_{cc}^3 + 576d_{bb}d_{cc}^2e_{ab}^2 - 1152d_{bb}d_{cc}e_{ab}e_{ac} + 1152d_{bb}d_{cc}e_{ab}e_{bc} + 576d_{bb}d_{cc}e_{ac}^2 - 1152d_{bb}d_{cc}e_{ac}e_{bc} + 576d_{bb}d_{cc}e_{bc}^2 + 240d_{cc}^4 + 288d_{cc}^2e_{ab}^2 - 576d_{cc}^2e_{ab}e_{ac} + 576d_{cc}^2e_{ab}e_{bc} + 288d_{cc}^2e_{ac}^2 - 576d_{cc}^2e_{ac}e_{bc} + 288d_{cc}^2e_{bc}^2 + 48e_{ab}^4 - 192e_{ab}^3e_{ac} + 192e_{ab}^3e_{bc} + 288e_{ab}^2e_{ac}^2 - 576e_{ab}^2e_{ac}e_{bc} + 288e_{ab}^2e_{bc}^2 - 192e_{ab}e_{ac}^3 + 576e_{ab}e_{ac}^2e_{bc} - 576e_{ab}e_{ac}e_{bc}^2 + 192e_{ab}e_{bc}^3 + 48e_{ac}^4 - 192e_{ac}^3e_{bc} + 288e_{ac}^2e_{bc}^2 - 192e_{ac}e_{bc}^3 + 48e_{bc}^4$

$ReE_{4,3}$ (153 terms; see the full text file)

$ReE_{4,6}$ (336 terms; see the full text file)

$ReE_{4,9}$ (225 terms; see the full text file)

$ReE_{4,12}$ (162 terms; see the full text file)

$ReE_{4,15}$ $162d_{ab}e_{ab}^2e_{ac} + 162d_{ab}e_{ab}^2e_{bc} + 162d_{ab}e_{ab}e_{ac}^2 - 162d_{ab}e_{ab}e_{bc}^2 - 162d_{ab}e_{ac}^2e_{bc} - 162d_{ab}e_{ac}e_{bc}^2 + 162d_{ac}e_{ab}^2e_{ac} + 162d_{ac}e_{ab}^2e_{bc} + 162d_{ac}e_{ab}e_{ac}^2 - 162d_{ac}e_{ab}e_{bc}^2 - 162d_{ac}e_{ac}^2e_{bc} - 162d_{ac}e_{ac}e_{bc}^2 + 162d_{bc}e_{ab}^2e_{ac} + 162d_{bc}e_{ab}^2e_{bc} + 162d_{bc}e_{ab}e_{ac}^2 - 162d_{bc}e_{ab}e_{bc}^2 - 162d_{bc}e_{ac}^2e_{bc} - 162d_{bc}e_{ac}e_{bc}^2$

$ReE_{5,0}$ (126 terms; see the full text file)

$ReE_{5,3}$ (429 terms; see the full text file)

$ReE_{5,6}$ (840 terms; see the full text file)

$ReE_{5,9}$ (837 terms; see the full text file)

$ReE_{5,12}$ (681 terms; see the full text file)

$ReE_{5,15}$ (318 terms; see the full text file)

$ReE_{5,18}$ $324d_{ab}^2e_{ab}^2e_{ac} + 324d_{ab}^2e_{ab}^2e_{bc} + 324d_{ab}^2e_{ab}e_{ac}^2 - 324d_{ab}^2e_{ab}e_{bc}^2 - 324d_{ab}^2e_{ac}^2e_{bc} - 324d_{ab}^2e_{ac}e_{bc}^2 + 648d_{ab}d_{ac}e_{ab}^2e_{ac} + 648d_{ab}d_{ac}e_{ab}^2e_{bc} + 648d_{ab}d_{ac}e_{ab}e_{ac}^2 - 648d_{ab}d_{ac}e_{ab}e_{bc}^2 - 648d_{ab}d_{ac}e_{ac}^2e_{bc} - 648d_{ab}d_{ac}e_{ac}e_{bc}^2 + 648d_{ab}d_{bc}e_{ab}^2e_{ac} + 648d_{ab}d_{bc}e_{ab}^2e_{bc} + 648d_{ab}d_{bc}e_{ab}e_{ac}^2 - 648d_{ab}d_{bc}e_{ab}e_{bc}^2 - 648d_{ab}d_{bc}e_{ac}^2e_{bc} - 648d_{ab}d_{bc}e_{ac}e_{bc}^2 - 54d_{ab}e_{ab}^4 - 108d_{ab}e_{ab}^3e_{ac} + 108d_{ab}e_{ab}^3e_{bc} - 81d_{ab}e_{ab}^2e_{ac}^2 + 162d_{ab}e_{ab}^2e_{ac}e_{bc} - 81d_{ab}e_{ab}^2e_{bc}^2 - 27d_{ab}e_{ab}e_{ac}^3 + 81d_{ab}e_{ab}e_{ac}^2e_{bc} - 81d_{ab}e_{ab}e_{ac}e_{bc}^2 + 27d_{ab}e_{ab}e_{bc}^3 + 27d_{ab}e_{ac}^4 + 135d_{ab}e_{ac}^3e_{bc} + 162d_{ab}e_{ac}^2e_{bc}^2 + 135d_{ab}e_{ac}e_{bc}^3 + 27d_{ab}e_{bc}^4 + 324d_{ac}^2e_{ab}^2e_{ac} + 324d_{ac}^2e_{ab}^2e_{bc} + 324d_{ac}^2e_{ab}e_{ac}^2 - 324d_{ac}^2e_{ab}e_{bc}^2 - 324d_{ac}^2e_{ac}^2e_{bc} - 324d_{ac}^2e_{ac}e_{bc}^2 + 648d_{ac}d_{bc}e_{ab}^2e_{ac} + 648d_{ac}d_{bc}e_{ab}^2e_{bc} + 648d_{ac}d_{bc}e_{ab}e_{ac}^2 - 648d_{ac}d_{bc}e_{ab}e_{bc}^2 - 648d_{ac}d_{bc}e_{ac}^2e_{bc} - 648d_{ac}d_{bc}e_{ac}e_{bc}^2 + 27d_{ac}e_{ab}^4 - 27d_{ac}e_{ab}^3e_{ac} - 135d_{ac}e_{ab}^3e_{bc} - 81d_{ac}e_{ab}^2e_{ac}^2 - 81d_{ac}e_{ab}^2e_{ac}e_{bc} + 162d_{ac}e_{ab}^2e_{bc}^2 - 108d_{ac}e_{ab}e_{ac}^3 - 162d_{ac}e_{ab}e_{ac}^2e_{bc} - 81d_{ac}e_{ab}e_{ac}e_{bc}^2 - 135d_{ac}e_{ab}e_{bc}^3 - 54d_{ac}e_{ac}^4 - 108d_{ac}e_{ac}^3e_{bc} - 81d_{ac}e_{ac}^2e_{bc}^2 - 27d_{ac}e_{ac}e_{bc}^3 + 27d_{ac}e_{bc}^4 + 324d_{bc}^2e_{ab}^2e_{ac} + 324d_{bc}^2e_{ab}^2e_{bc} + 324d_{bc}^2e_{ab}e_{ac}^2 - 324d_{bc}^2e_{ab}e_{bc}^2 - 324d_{bc}^2e_{ac}^2e_{bc} - 324d_{bc}^2e_{ac}e_{bc}^2 + 27d_{bc}e_{ab}^4 + 135d_{bc}e_{ab}^3e_{ac} + 27d_{bc}e_{ab}^3e_{bc} + 162d_{bc}e_{ab}^2e_{ac}^2 - 81d_{bc}e_{ab}^2e_{ac}e_{bc} - 81d_{bc}e_{ab}^2e_{bc}^2 + 135d_{bc}e_{ab}e_{ac}^3 + 81d_{bc}e_{ab}e_{ac}^2e_{bc} + 162d_{bc}e_{ab}e_{ac}e_{bc}^2 + 108d_{bc}e_{ab}e_{bc}^3 + 27d_{bc}e_{ac}^4 - 27d_{bc}e_{ac}^3e_{bc} - 81d_{bc}e_{ac}^2e_{bc}^2 - 108d_{bc}e_{ac}e_{bc}^3 - 54d_{bc}e_{bc}^4$

$ReE_{6,0}$ (236 terms; see the full text file)

$ReE_{6,3}$ (820 terms; see the full text file)

$ReE_{6,6}$ (2213 terms; see the full text file)

$ReE_{6,9}$ (2302 terms; see the full text file)

$ReE_{6,12}$ (2391 terms; see the full text file)

$ReE_{6,15}$ (1414 terms; see the full text file)

$ReE_{6,18}$ (913 terms; see the full text file)

$ReE_{6,21}$ (204 terms; see the full text file)

$ReE_{6,24}$ $27e_{ab}^6 + 81e_{ab}^5e_{ac} - 81e_{ab}^5e_{bc} + 162e_{ab}^4e_{ac}^2 - 81e_{ab}^4e_{ac}e_{bc} + 162e_{ab}^4e_{bc}^2 + 189e_{ab}^3e_{ac}^3 - 81e_{ab}^3e_{ac}^2e_{bc} + 81e_{ab}^3e_{ac}e_{bc}^2 - 189e_{ab}^3e_{bc}^3 + 162e_{ab}^2e_{ac}^4 + 81e_{ab}^2e_{ac}^3e_{bc} + 243e_{ab}^2e_{ac}^2e_{bc}^2 + 81e_{ab}^2e_{ac}e_{bc}^3 + 162e_{ab}^2e_{bc}^4 + 81e_{ab}e_{ac}^5 + 81e_{ab}e_{ac}^4e_{bc} + 81e_{ab}e_{ac}^3e_{bc}^2 - 81e_{ab}e_{ac}^2e_{bc}^3 - 81e_{ab}e_{ac}e_{bc}^4 - 81e_{ab}e_{bc}^5 + 27e_{ac}^6 + 81e_{ac}^5e_{bc} + 162e_{ac}^4e_{bc}^2 + 189e_{ac}^3e_{bc}^3 + 162e_{ac}^2e_{bc}^4 + 81e_{ac}e_{bc}^5 + 27e_{bc}^6$

3 $S(2) \times S(1)$ acting on (a, b, p), $k = 2$

$|G| = 2$, $m = 9$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 11 (plain Bursarial: 26; quotient dimension 5).
Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{pp}$
- inv[2] $= 2d_{ab}$
- inv[3] $= d_{ap} + d_{bp}$
- inv[4] $= e_{ap} + e_{bp}$

j	weight	real part of z_j	imaginary part of z_j
0	1	$d_{aa} - d_{bb}$	$2e_{ab}$
1	1	$d_{ap} - d_{bp}$	$e_{ap} - e_{bp}$

output	expanded
$inv[0]$	$d_{aa} + d_{bb}$
$inv[1]$	$2d_{pp}$
$inv[2]$	$2d_{ab}$
$inv[3]$	$d_{ap} + d_{bp}$
$inv[4]$	$e_{ap} + e_{bp}$
$ReE_{2,0}$	$-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + 4e_{ab}^2$
$ImE_{2,0}$	$-4d_{aa}e_{ab} + 4d_{bb}e_{ab}$
$ReE_{2,1}$	$-2d_{aa}d_{ap} + 2d_{aa}d_{bp} + 2d_{ap}d_{bb} - 2d_{bb}d_{bp} + 4e_{ab}e_{ap} - 4e_{ab}e_{bp}$
$ImE_{2,1}$	$-2d_{aa}e_{ap} + 2d_{aa}e_{bp} - 4d_{ap}e_{ab} + 2d_{bb}e_{ap} - 2d_{bb}e_{bp} + 4d_{bp}e_{ab}$
$ReE_{2,2}$	$-d_{ap}^2 + 2d_{ap}d_{bp} - d_{bp}^2 + e_{ap}^2 - 2e_{ap}e_{bp} + e_{bp}^2$
$ImE_{2,2}$	$-2d_{ap}e_{ap} + 2d_{ap}e_{bp} + 2d_{bp}e_{ap} - 2d_{bp}e_{bp}$

4 $S(2)$ acting on (a, b), $k = 1$

$|G| = 2$, $m = 5$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 5 (plain Bursarial: 14; quotient dimension 2).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{ab}$
- inv[2] $= e_a + e_b$

j	weight	real part of z_j	imaginary part of z_j
0	1	$d_{aa} - d_{bb}$	$e_a - e_b$

output	expanded
$inv[0]$	$d_{aa} + d_{bb}$
$inv[1]$	$2d_{ab}$
$inv[2]$	$e_a + e_b$
$ReE_{2,0}$	$-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + e_a^2 - 2e_ae_b + e_b^2$
$ImE_{2,0}$	$-2d_{aa}e_a + 2d_{aa}e_b + 2d_{bb}e_a - 2d_{bb}e_b$

5 $S(3)$ acting on (a, b, c), $k = 1$

$|G| = 6$, $m = 9$ slots; $h = (abc)$ (order $r = 3$), $s = (bc)$, mode x; real coefficients. Outputs: 24 (plain Bursarial: 174; quotient dimension 3).

Invariants output directly:

- inv[0] $= 2d_{aa} + 2d_{bb} + 2d_{cc}$
- inv[1] $= 2d_{ab} + 2d_{ac} + 2d_{bc}$
- inv[2] $= 2e_a + 2e_b + 2e_c$

j	weight	real part of z_j	imaginary part of z_j
0		0	0
1	1	$2d_{aa} - d_{bb} - d_{cc}$	$\sqrt{3}\,(-d_{bb} + d_{cc})$
2	2	$\frac{d_{ab}}{2} + \frac{d_{ac}}{2} - d_{bc}$	$\frac{\sqrt{3}(d_{ab} - d_{ac})}{2}$
3		0	0
4	1	$2e_a - e_b - e_c$	$\sqrt{3}\,(-e_b + e_c)$

output	expanded
$inv[0]$	$2d_{aa} + 2d_{bb} + 2d_{cc}$
$inv[1]$	$2d_{ab} + 2d_{ac} + 2d_{bc}$
$inv[2]$	$2e_a + 2e_b + 2e_c$
$ReE_{2,3}$	$-6d_{aa}d_{ab} - 6d_{aa}d_{ac} + 12d_{aa}d_{bc} - 6d_{ab}d_{bb} + 12d_{ab}d_{cc} + 12d_{ac}d_{bb} - 6d_{ac}d_{cc} - 6d_{bb}d_{bc} - 6d_{bc}d_{cc}$
$ReE_{2,6}$	$-6d_{ab}e_a - 6d_{ab}e_b + 12d_{ab}e_c - 6d_{ac}e_a + 12d_{ac}e_b - 6d_{ac}e_c + 12d_{bc}e_a - 6d_{bc}e_b - 6d_{bc}e_c$
$ReE_{3,3}$	$16d_{aa}^3 - 24d_{aa}^2d_{bb} - 24d_{aa}^2d_{cc} - 24d_{aa}d_{bb}^2 + 96d_{aa}d_{bb}d_{cc} - 24d_{aa}d_{cc}^2 + 16d_{bb}^3 - 24d_{bb}^2d_{cc} - 24d_{bb}d_{cc}^2 + 16d_{cc}^3$
$ReE_{3,6}$	$48d_{aa}^2e_a - 24d_{aa}^2e_b - 24d_{aa}^2e_c - 48d_{aa}d_{bb}e_a - 48d_{aa}d_{bb}e_b + 96d_{aa}d_{bb}e_c - 48d_{aa}d_{cc}e_a + 96d_{aa}d_{cc}e_b - 48d_{aa}d_{cc}e_c - 2d_{ab}^3 + 3d_{ab}^2d_{ac} + 3d_{ab}^2d_{bc} + 3d_{ab}d_{ac}^2 - 12d_{ab}d_{ac}d_{bc} + 3d_{ab}d_{bc}^2 - 2d_{ac}^3 + 3d_{ac}^2d_{bc} + 3d_{ac}d_{bc}^2 - 24d_{bb}^2e_a + 48d_{bb}^2e_b - 24d_{bb}^2e_c + 96d_{bb}d_{cc}e_a - 48d_{bb}d_{cc}e_b - 48d_{bb}d_{cc}e_c - 2d_{bc}^3 - 24d_{cc}^2e_a - 24d_{cc}^2e_b + 48d_{cc}^2e_c$
$ReE_{3,9}$	$48d_{aa}e_a^2 - 48d_{aa}e_ae_b - 48d_{aa}e_ae_c - 24d_{aa}e_b^2 + 96d_{aa}e_be_c - 24d_{aa}e_c^2 - 24d_{bb}e_a^2 - 48d_{bb}e_ae_b + 96d_{bb}e_ae_c + 48d_{bb}e_b^2 - 48d_{bb}e_be_c - 24d_{bb}e_c^2 - 24d_{cc}e_a^2 + 96d_{cc}e_ae_b - 48d_{cc}e_ae_c - 24d_{cc}e_b^2 - 48d_{cc}e_be_c + 48d_{cc}e_c^2$
$ReE_{3,12}$	$16e_a^3 - 24e_a^2e_b - 24e_a^2e_c - 24e_ae_b^2 + 96e_ae_be_c - 24e_ae_c^2 + 16e_b^3 - 24e_b^2e_c - 24e_be_c^2 + 16e_c^3$
$ReE_{4,6}$	$36d_{aa}^2d_{ab}^2 - 36d_{aa}^2d_{ab}d_{ac} - 36d_{aa}^2d_{ab}d_{bc} + 36d_{aa}^2d_{ac}^2 - 36d_{aa}^2d_{ac}d_{bc} + 36d_{aa}^2d_{bc}^2 - 36d_{aa}d_{ab}^2d_{bb} - 36d_{aa}d_{ab}^2d_{cc} + 36d_{aa}d_{ab}d_{ac}d_{bb} + 36d_{aa}d_{ab}d_{ac}d_{cc} + 36d_{aa}d_{ab}d_{bb}d_{bc} + 36d_{aa}d_{ab}d_{bc}d_{cc} - 36d_{aa}d_{ac}^2d_{bb} - 36d_{aa}d_{ac}^2d_{cc} + 36d_{aa}d_{ac}d_{bb}d_{bc} + 36d_{aa}d_{ac}d_{bc}d_{cc} - 36d_{aa}d_{bb}d_{bc}^2 - 36d_{aa}d_{bc}^2d_{cc} + 36d_{ab}^2d_{bb}^2 - 36d_{ab}^2d_{bb}d_{cc} + 36d_{ab}^2d_{cc}^2 - 36d_{ab}d_{ac}d_{bb}^2 + 36d_{ab}d_{ac}d_{bb}d_{cc} - 36d_{ab}d_{ac}d_{cc}^2 - 36d_{ab}d_{bb}^2d_{bc} + 36d_{ab}d_{bb}d_{bc}d_{cc} - 36d_{ab}d_{bc}d_{cc}^2 + 36d_{ac}^2d_{bb}^2 - 36d_{ac}^2d_{bb}d_{cc} + 36d_{ac}^2d_{cc}^2 - 36d_{ac}d_{bb}^2d_{bc} + 36d_{ac}d_{bb}d_{bc}d_{cc} - 36d_{ac}d_{bc}d_{cc}^2 + 36d_{bb}^2d_{bc}^2 - 36d_{bb}d_{bc}^2d_{cc} + 36d_{bc}^2d_{cc}^2$

$ReE_{4,9}$ $72d_{aa}d_{ab}^2e_a - 36d_{aa}d_{ab}^2e_b - 36d_{aa}d_{ab}^2e_c - 72d_{aa}d_{ab}d_{ac}e_a + 36d_{aa}d_{ab}d_{ac}e_b + 36d_{aa}d_{ab}d_{ac}e_c - 72d_{aa}d_{ab}d_{bc}e_a + 36d_{aa}d_{ab}d_{bc}e_b + 36d_{aa}d_{ab}d_{bc}e_c + 72d_{aa}d_{ac}^2e_a - 36d_{aa}d_{ac}^2e_b - 36d_{aa}d_{ac}^2e_c - 72d_{aa}d_{ac}d_{bc}e_a + 36d_{aa}d_{ac}d_{bc}e_b + 36d_{aa}d_{ac}d_{bc}e_c + 72d_{aa}d_{bc}^2e_a - 36d_{aa}d_{bc}^2e_b - 36d_{aa}d_{bc}^2e_c - 36d_{ab}^2d_{bb}e_a + 72d_{ab}^2d_{bb}e_b - 36d_{ab}^2d_{bb}e_c - 36d_{ab}^2d_{cc}e_a - 36d_{ab}^2d_{cc}e_b + 72d_{ab}^2d_{cc}e_c + 36d_{ab}d_{ac}d_{bb}e_a - 72d_{ab}d_{ac}d_{bb}e_b + 36d_{ab}d_{ac}d_{bb}e_c + 36d_{ab}d_{ac}d_{cc}e_a + 36d_{ab}d_{ac}d_{cc}e_b - 72d_{ab}d_{ac}d_{cc}e_c + 36d_{ab}d_{bb}d_{bc}e_a - 72d_{ab}d_{bb}d_{bc}e_b + 36d_{ab}d_{bb}d_{bc}e_c + 36d_{ab}d_{bc}d_{cc}e_a + 36d_{ab}d_{bc}d_{cc}e_b - 72d_{ab}d_{bc}d_{cc}e_c - 36d_{ac}^2d_{bb}e_a + 72d_{ac}^2d_{bb}e_b - 36d_{ac}^2d_{bb}e_c - 36d_{ac}^2d_{cc}e_a - 36d_{ac}^2d_{cc}e_b + 72d_{ac}^2d_{cc}e_c + 36d_{ac}d_{bb}d_{bc}e_a - 72d_{ac}d_{bb}d_{bc}e_b + 36d_{ac}d_{bb}d_{bc}e_c + 36d_{ac}d_{bc}d_{cc}e_a + 36d_{ac}d_{bc}d_{cc}e_b - 72d_{ac}d_{bc}d_{cc}e_c - 36d_{bb}d_{bc}^2e_a + 72d_{bb}d_{bc}^2e_b - 36d_{bb}d_{bc}^2e_c - 36d_{bc}^2d_{cc}e_a - 36d_{bc}^2d_{cc}e_b + 72d_{bc}^2d_{cc}e_c$

$ReE_{4,12}$ $36d_{ab}^2e_a^2 - 36d_{ab}^2e_ae_b - 36d_{ab}^2e_ae_c + 36d_{ab}^2e_b^2 - 36d_{ab}^2e_be_c + 36d_{ab}^2e_c^2 - 36d_{ab}d_{ac}e_a^2 + 36d_{ab}d_{ac}e_ae_b + 36d_{ab}d_{ac}e_ae_c - 36d_{ab}d_{ac}e_b^2 + 36d_{ab}d_{ac}e_be_c - 36d_{ab}d_{ac}e_c^2 - 36d_{ab}d_{bc}e_a^2 + 36d_{ab}d_{bc}e_ae_b + 36d_{ab}d_{bc}e_ae_c - 36d_{ab}d_{bc}e_b^2 + 36d_{ab}d_{bc}e_be_c - 36d_{ab}d_{bc}e_c^2 + 36d_{ac}^2e_a^2 - 36d_{ac}^2e_ae_b - 36d_{ac}^2e_ae_c + 36d_{ac}^2e_b^2 - 36d_{ac}^2e_be_c + 36d_{ac}^2e_c^2 - 36d_{ac}d_{bc}e_a^2 + 36d_{ac}d_{bc}e_ae_b + 36d_{ac}d_{bc}e_ae_c - 36d_{ac}d_{bc}e_b^2 + 36d_{ac}d_{bc}e_be_c - 36d_{ac}d_{bc}e_c^2 + 36d_{bc}^2e_a^2 - 36d_{bc}^2e_ae_b - 36d_{bc}^2e_ae_c + 36d_{bc}^2e_b^2 - 36d_{bc}^2e_be_c + 36d_{bc}^2e_c^2$

$ReE_{5,6}$ $-48d_{aa}^4d_{ab} - 48d_{aa}^4d_{ac} + 96d_{aa}^4d_{bc} + 240d_{aa}^3d_{ab}d_{bb} - 48d_{aa}^3d_{ab}d_{cc} - 48d_{aa}^3d_{ac}d_{bb} + 240d_{aa}^3d_{ac}d_{cc} - 192d_{aa}^3d_{bb}d_{bc} - 192d_{aa}^3d_{bc}d_{cc} - 288d_{aa}^2d_{ab}d_{bb}^2 - 144d_{aa}^2d_{ab}d_{bb}d_{cc} + 144d_{aa}^2d_{ab}d_{cc}^2 + 144d_{aa}^2d_{ac}d_{bb}^2 - 144d_{aa}^2d_{ac}d_{bb}d_{cc} - 288d_{aa}^2d_{ac}d_{cc}^2 + 144d_{aa}^2d_{bb}^2d_{bc} + 288d_{aa}^2d_{bb}d_{bc}d_{cc} + 144d_{aa}^2d_{bc}d_{cc}^2 + 240d_{aa}d_{ab}d_{bb}^3 - 144d_{aa}d_{ab}d_{bb}^2d_{cc} + 288d_{aa}d_{ab}d_{bb}d_{cc}^2 - 192d_{aa}d_{ab}d_{cc}^3 - 192d_{aa}d_{ac}d_{bb}^3 + 288d_{aa}d_{ac}d_{bb}^2d_{cc} - 144d_{aa}d_{ac}d_{bb}d_{cc}^2 + 240d_{aa}d_{ac}d_{cc}^3 - 48d_{aa}d_{bb}^3d_{bc} - 144d_{aa}d_{bb}^2d_{bc}d_{cc} - 144d_{aa}d_{bb}d_{bc}d_{cc}^2 - 48d_{aa}d_{bc}d_{cc}^3 - 48d_{ab}d_{bb}^4 - 48d_{ab}d_{bb}^3d_{cc} + 144d_{ab}d_{bb}^2d_{cc}^2 - 192d_{ab}d_{bb}d_{cc}^3 + 96d_{ab}d_{cc}^4 + 96d_{ac}d_{bb}^4 - 192d_{ac}d_{bb}^3d_{cc} + 144d_{ac}d_{bb}^2d_{cc}^2 - 48d_{ac}d_{bb}d_{cc}^3 - 48d_{ac}d_{cc}^4 - 48d_{bb}^4d_{bc} + 240d_{bb}^3d_{bc}d_{cc} - 288d_{bb}^2d_{bc}d_{cc}^2 + 240d_{bb}d_{bc}d_{cc}^3 - 48d_{bc}d_{cc}^4$

$ReE_{5,9}$ (135 terms; see the full text file)

$ReE_{5,12}$ (153 terms; see the full text file)

$ReE_{5,15}$ $-192d_{aa}d_{ab}e_a^3 + 720d_{aa}d_{ab}e_a^2e_b - 144d_{aa}d_{ab}e_a^2e_c - 576d_{aa}d_{ab}e_ae_b^2 - 288d_{aa}d_{ab}e_ae_be_c + 288d_{aa}d_{ab}e_ae_c^2 + 240d_{aa}d_{ab}e_b^3 - 144d_{aa}d_{ab}e_b^2e_c + 288d_{aa}d_{ab}e_be_c^2 - 192d_{aa}d_{ab}e_c^3 - 192d_{aa}d_{ac}e_a^3 - 144d_{aa}d_{ac}e_a^2e_b + 720d_{aa}d_{ac}e_a^2e_c + 288d_{aa}d_{ac}e_ae_b^2 - 288d_{aa}d_{ac}e_ae_be_c - 576d_{aa}d_{ac}e_ae_c^2 - 192d_{aa}d_{ac}e_b^3 + 288d_{aa}d_{ac}e_b^2e_c - 144d_{aa}d_{ac}e_be_c^2 + 240d_{aa}d_{ac}e_c^3 + 384d_{aa}d_{bc}e_a^3 - 576d_{aa}d_{bc}e_a^2e_b - 576d_{aa}d_{bc}e_a^2e_c + 288d_{aa}d_{bc}e_ae_b^2 + 576d_{aa}d_{bc}e_ae_be_c + 288d_{aa}d_{bc}e_ae_c^2 - 48d_{aa}d_{bc}e_b^3 - 144d_{aa}d_{bc}e_b^2e_c - 144d_{aa}d_{bc}e_be_c^2 - 48d_{aa}d_{bc}e_c^3 + 240d_{ab}d_{bb}e_a^3 - 576d_{ab}d_{bb}e_a^2e_b - 144d_{ab}d_{bb}e_a^2e_c + 720d_{ab}d_{bb}e_ae_b^2 - 288d_{ab}d_{bb}e_ae_be_c + 288d_{ab}d_{bb}e_ae_c^2 - 192d_{ab}d_{bb}e_b^3 - 144d_{ab}d_{bb}e_b^2e_c + 288d_{ab}d_{bb}e_be_c^2 - 192d_{ab}d_{bb}e_c^3 - 48d_{ab}d_{cc}e_a^3 - 144d_{ab}d_{cc}e_a^2e_b + 288d_{ab}d_{cc}e_a^2e_c - 144d_{ab}d_{cc}e_ae_b^2 + 576d_{ab}d_{cc}e_ae_be_c - 576d_{ab}d_{cc}e_ae_c^2 - 48d_{ab}d_{cc}e_b^3 + 288d_{ab}d_{cc}e_b^2e_c - 576d_{ab}d_{cc}e_be_c^2 + 384d_{ab}d_{cc}e_c^3 - 48d_{ac}d_{bb}e_a^3 + 288d_{ac}d_{bb}e_a^2e_b - 144d_{ac}d_{bb}e_a^2e_c - 576d_{ac}d_{bb}e_ae_be_c + 576d_{ac}d_{bb}e_ae_be_c - 144d_{ac}d_{bb}e_ae_c^2 + 384d_{ac}d_{bb}e_b^3 - 576d_{ac}d_{bb}e_b^2e_c + 288d_{ac}d_{bb}e_be_c^2 - 48d_{ac}d_{bb}e_c^3 + 240d_{ac}d_{cc}e_a^3 - 144d_{ac}d_{cc}e_a^2e_b - 576d_{ac}d_{cc}e_a^2e_c + 288d_{ac}d_{cc}e_ae_b^2 - 288d_{ac}d_{cc}e_ae_be_c + 720d_{ac}d_{cc}e_ae_c^2 - 192d_{ac}d_{cc}e_b^3 + 288d_{ac}d_{cc}e_b^2e_c - 144d_{ac}d_{cc}e_be_c^2 - 192d_{ac}d_{cc}e_c^3 - 192d_{bb}d_{bc}e_a^3 + 288d_{bb}d_{bc}e_a^2e_b + 288d_{bb}d_{bc}e_a^2e_c - 144d_{bb}d_{bc}e_ae_b^2 - 288d_{bb}d_{bc}e_ae_be_c - 144d_{bb}d_{bc}e_ae_c^2 - 192d_{bb}d_{bc}e_b^3 + 720d_{bb}d_{bc}e_b^2e_c - 576d_{bb}d_{bc}e_be_c^2 + 240d_{bb}d_{bc}e_c^3 - 192d_{bc}d_{cc}e_a^3 + 288d_{bc}d_{cc}e_a^2e_b + 288d_{bc}d_{cc}e_a^2e_c - 144d_{bc}d_{cc}e_ae_b^2 - 288d_{bc}d_{cc}e_ae_be_c - 144d_{bc}d_{cc}e_ae_c^2 + 240d_{bc}d_{cc}e_b^3 - 576d_{bc}d_{cc}e_b^2e_c + 720d_{bc}d_{cc}e_be_c^2 - 192d_{bc}d_{cc}e_c^3$

$ReE_{5,18}$ $-48d_{ab}e_a^4 + 240d_{ab}e_a^3e_b - 48d_{ab}e_a^3e_c - 288d_{ab}e_a^2e_b^2 - 144d_{ab}e_a^2e_be_c + 144d_{ab}e_a^2e_c^2 + 240d_{ab}e_ae_b^3 - 144d_{ab}e_ae_b^2e_c + 288d_{ab}e_ae_be_c^2 - 192d_{ab}e_ae_c^3 - 48d_{ab}e_b^4 - 48d_{ab}e_b^3e_c + 144d_{ab}e_b^2e_c^2 - 192d_{ab}e_be_c^3 + 96d_{ab}e_c^4 - 48d_{ac}e_a^4 - 48d_{ac}e_a^3e_b + 240d_{ac}e_a^3e_c + 144d_{ac}e_a^2e_b^2 - 144d_{ac}e_a^2e_be_c - 288d_{ac}e_a^2e_c^2 - 192d_{ac}e_ae_b^3 + 288d_{ac}e_ae_b^2e_c - 144d_{ac}e_ae_be_c^2 + 240d_{ac}e_ae_c^3 + 96d_{ac}e_b^4 - 192d_{ac}e_b^3e_c + 144d_{ac}e_b^2e_c^2 - 48d_{ac}e_be_c^3 - 48d_{ac}e_c^4 + 96d_{bc}e_a^4 - 192d_{bc}e_a^3e_b - 192d_{bc}e_a^3e_c + 144d_{bc}e_a^2e_b^2 + 288d_{bc}e_a^2e_be_c + 144d_{bc}e_a^2e_c^2 - 48d_{bc}e_ae_b^3 - 144d_{bc}e_ae_b^2e_c - 144d_{bc}e_ae_be_c^2 - 48d_{bc}e_ae_c^3 - 48d_{bc}e_b^4 + 240d_{bc}e_b^3e_c - 288d_{bc}e_b^2e_c^2 + 240d_{bc}e_be_c^3 - 48d_{bc}e_c^4$

$ReE_{6,6}$ $64d_{aa}^6 - 192d_{aa}^5d_{bb} - 192d_{aa}^5d_{cc} + 384d_{aa}^4d_{bb}^2 + 192d_{aa}^4d_{bb}d_{cc} + 384d_{aa}^4d_{cc}^2 - 448d_{aa}^3d_{bb}^3 - 192d_{aa}^3d_{bb}^2d_{cc} - 192d_{aa}^3d_{bb}d_{cc}^2 - 448d_{aa}^3d_{cc}^3 + 384d_{aa}^2d_{bb}^4 - 192d_{aa}^2d_{bb}^3d_{cc} + 576d_{aa}^2d_{bb}^2d_{cc}^2 - 192d_{aa}^2d_{bb}d_{cc}^3 + 384d_{aa}^2d_{cc}^4 - 192d_{aa}d_{bb}^5 + 192d_{aa}d_{bb}^4d_{cc} - 192d_{aa}d_{bb}^3d_{cc}^2 - 192d_{aa}d_{bb}^2d_{cc}^3 + 192d_{aa}d_{bb}d_{cc}^4 - 192d_{aa}d_{cc}^5 + 64d_{bb}^6 - 192d_{bb}^5d_{cc} + 384d_{bb}^4d_{cc}^2 - 448d_{bb}^3d_{cc}^3 + 384d_{bb}^2d_{cc}^4 - 192d_{bb}d_{cc}^5 + 64d_{cc}^6$

$ReE_{6,9}$ (163 terms; see the full text file)

$ReE_{6,12}$ (298 terms; see the full text file)

$ReE_{6,15}$ (280 terms; see the full text file)

$ReE_{6,18}$ (190 terms; see the full text file)

$ReE_{6,21}$ $384d_{aa}e_a^5 - 960d_{aa}e_a^4e_b - 960d_{aa}e_a^4e_c + 1536d_{aa}e_a^3e_b^2 + 768d_{aa}e_a^3e_be_c + 1536d_{aa}e_a^3e_c^2 - 1344d_{aa}e_a^2e_b^3 - 576d_{aa}e_a^2e_b^2e_c - 576d_{aa}e_a^2e_be_c^2 - 1344d_{aa}e_a^2e_c^3 + 768d_{aa}e_ae_b^4 - 384d_{aa}e_ae_b^3e_c + 1152d_{aa}e_ae_b^2e_c^2 - 384d_{aa}e_ae_be_c^3 + 768d_{aa}e_ae_c^4 - 192d_{aa}e_b^5 + 192d_{aa}e_b^4e_c - 192d_{aa}e_b^3e_c^2 - 192d_{aa}e_b^2e_c^3 + 192d_{aa}e_be_c^4 - 192d_{aa}e_c^5 - 192d_{bb}e_a^5 + 768d_{bb}e_a^4e_b + 192d_{bb}e_a^4e_c - 1344d_{bb}e_a^3e_b^2 - 384d_{bb}e_a^3e_be_c - 192d_{bb}e_a^3e_c^2 + 1536d_{bb}e_a^2e_b^3 - 576d_{bb}e_a^2e_b^2e_c + 1152d_{bb}e_a^2e_be_c^2 - 192d_{bb}e_a^2e_c^3 - 960d_{bb}e_ae_b^4 + 768d_{bb}e_ae_b^3e_c - 576d_{bb}e_ae_b^2e_c^2 - 384d_{bb}e_ae_be_c^3 + 192d_{bb}e_ae_c^4 + 384d_{bb}e_b^5 - 960d_{bb}e_b^4e_c + 1536d_{bb}e_b^3e_c^2 - 1344d_{bb}e_b^2e_c^3 + 768d_{bb}e_be_c^4 - 192d_{bb}e_c^5 - 192d_{cc}e_a^5 + 192d_{cc}e_a^4e_b + 768d_{cc}e_a^4e_c - 192d_{cc}e_a^3e_b^2 - 384d_{cc}e_a^3e_be_c - 1344d_{cc}e_a^3e_c^2 - 192d_{cc}e_a^2e_b^3 + 1152d_{cc}e_a^2e_b^2e_c - 576d_{cc}e_a^2e_be_c^2 + 1536d_{cc}e_a^2e_c^3 + 192d_{cc}e_ae_b^4 - 384d_{cc}e_ae_b^3e_c - 576d_{cc}e_ae_b^2e_c^2 + 768d_{cc}e_ae_be_c^3 - 960d_{cc}e_ae_c^4 - 192d_{cc}e_b^5 + 768d_{cc}e_b^4e_c - 1344d_{cc}e_b^3e_c^2 + 1536d_{cc}e_b^2e_c^3 - 960d_{cc}e_be_c^4 + 384d_{cc}e_c^5$

$ReE_{6,24}$ $64e_a^6 - 192e_a^5e_b - 192e_a^5e_c + 384e_a^4e_b^2 + 192e_a^4e_be_c + 384e_a^4e_c^2 - 448e_a^3e_b^3 - 192e_a^3e_b^2e_c - 192e_a^3e_be_c^2 - 448e_a^3e_c^3 + 384e_a^2e_b^4 - 192e_a^2e_b^3e_c + 576e_a^2e_b^2e_c^2 - 192e_a^2e_be_c^3 + 384e_a^2e_c^4 - 192e_ae_b^5 + 192e_ae_b^4e_c - 192e_ae_b^3e_c^2 - 192e_ae_b^2e_c^3 + 192e_ae_be_c^4 - 192e_ae_c^5 + 64e_b^6 - 192e_b^5e_c + 384e_b^4e_c^2 - 448e_b^3e_c^3 + 384e_b^2e_c^4 - 192e_be_c^5 + 64e_c^6$

6 $S(2) \times S(1)$ acting on (a, b, p), $k = 1$

$|G| = 2$, $m = 9$ slots; $h = (ab)$ (order $r = 2$), $s = e$, mode scalar; real coefficients. Outputs: 11 (plain Bursarial: 26; quotient dimension 3).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{pp}$
- inv[2] $= 2d_{ab}$
- inv[3] $= d_{ap} + d_{bp}$
- inv[4] $= e_a + e_b$
- inv[5] $= 2e_p$

j	weight	real part of z_j	imaginary part of z_j
0	1	$2d_{aa} - 2d_{bb}$	0
1	1	$2d_{ap} - 2d_{bp}$	0
2	1	$2e_a - 2e_b$	0

output	expanded
$inv[0]$	$d_{aa} + d_{bb}$
$inv[1]$	$2d_{pp}$
$inv[2]$	$2d_{ab}$
$inv[3]$	$d_{ap} + d_{bp}$
$inv[4]$	$e_a + e_b$
$inv[5]$	$2e_p$
$ReE_{2,0}$	$-4d_{aa}^2 + 8d_{aa}d_{bb} - 4d_{bb}^2$
$ReE_{2,1}$	$-8d_{aa}d_{ap} + 8d_{aa}d_{bp} + 8d_{ap}d_{bb} - 8d_{bb}d_{bp}$
$ReE_{2,2}$	$-8d_{aa}e_a + 8d_{aa}e_b - 4d_{ap}^2 + 8d_{ap}d_{bp} + 8d_{bb}e_a - 8d_{bb}e_b - 4d_{bp}^2$
$ReE_{2,3}$	$-8d_{ap}e_a + 8d_{ap}e_b + 8d_{bp}e_a - 8d_{bp}e_b$
$ReE_{2,4}$	$-4e_a^2 + 8e_ae_b - 4e_b^2$